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Transformer-Based Review Understanding with RAG-Enhanced Explanation Generation

1 Introduction

The objective of this assignment is to design and implement a complete Retrieval-Augmented Generation (RAG) pipeline for review understanding and explanation generation. The system comprises three integrated stages: an encoder-only Transformer that performs multi-task learning to extract sentiment and category information from product reviews while producing fixed-dimensional vector representations; a retrieval module that indexes these representations and retrieves semantically similar reviews from the training corpus; and a decoder-only Transformer that generates natural language explanations conditioned on the original review, predicted labels, and retrieved similar examples. The entire pipeline is trained on a self-constructed subset of the Amazon Reviews dataset containing 36,000 reviews across three categories: Cellphones, Electronics, and Sports.

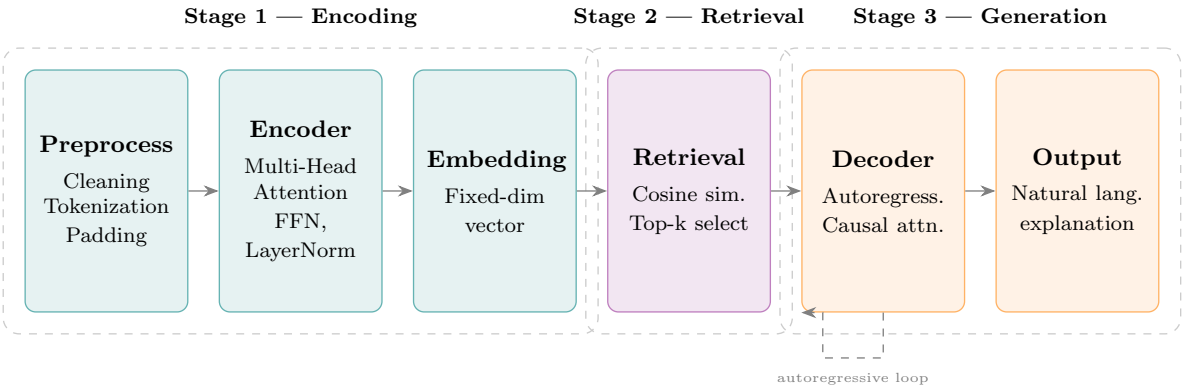
2 System Design and Methodology

The pipeline follows a sequential architecture. Raw review text is preprocessed through lowercasing, removal of HTML/URLs, filtering of non-alphabetic characters, and whitespace normalization. The cleaned text is tokenized, and a vocabulary is built from the training split using a minimum frequency threshold of two. Reviews are encoded into fixed-length sequences of size 128 using right-truncation and padding.

An encoder-only Transformer with multi-head self-attention, feed-forward layers, residual connections, and layer normalization processes the input. It outputs sentiment logits (Negative, Neutral, Positive), category logits (Cellphones, Electronics, Sports), and a fixed-dimensional embedding via mean pooling over valid tokens. Training is performed end-to-end using a joint loss with equal weighting of the two cross-entropy objectives.

At inference, embeddings for test reviews are generated and compared against precomputed, normalized training embeddings using cosine similarity. The top-k similar samples are retrieved and formatted into a structured context.

A decoder-only Transformer generates explanations autoregressively. Its input combines the review text, predicted sentiment and category labels, retrieved context, and a generation token. The model is trained with cross-entropy on target explanation sequences. During inference, tokens are generated using temperature-controlled sampling until an end-of-sequence token or length limit is reached.



3 Design Decisions and Justification

The encoder architecture employs a 3-layer Transformer with 128-dimensional embeddings and 4 attention heads. This configuration balances representational capacity with computational efficiency given the dataset size of 36,000 samples. The feed-forward dimension is set to 256, following the typical practice of using a hidden dimension two to four times the model dimension. Dropout of 0.2 is applied throughout to mitigate overfitting, as the initial value of 0.1 led to training-validation divergence. The Adam optimizer with a learning rate of 0.0003, beta parameters of (0.9, 0.98), and epsilon of 1e-9 is used, combined with a warmup-cosine learning rate scheduler that linearly increases the learning rate for the first 500 steps before decaying with a cosine schedule.

For the decoder, a larger architecture with 256-dimensional embeddings, 8 attention heads, 4 layers, and a feed-forward dimension of 512 is chosen to provide sufficient capacity for language generation. The decoder is trained with a batch size of 8 and gradient accumulation over 2 steps to achieve an effective batch size of 16 within GPU memory constraints. Mixed precision training is enabled using automatic mixed precision to accelerate training and reduce

5 Experimental Setup

The Amazon Reviews dataset is sampled with 12,000 reviews from each of three categories: Cellphones, Electronics, and Sports, resulting in a total of 36,000 reviews. The data is shuffled and split into training, validation, and test sets with ratios of 70%, 15%, and 15%, respectively, yielding 25,200 training samples, 5,400 validation samples, and 5,400 test samples. The encoder is trained for a maximum of 20 epochs with early stopping patience of 5 epochs based on validation loss. The training uses batch sizes of 128 for the encoder and 8 for the decoder. The encoder loss combines sentiment and category cross-entropy with equal weighting. The decoder loss is computed only over positions after the explanation token using a mask, ignoring padding tokens. All experiments are conducted on a Tesla T4 GPU with 15.6 GB of VRAM using PyTorch as the deep learning framework.

6 Evaluation Results

The encoder achieves strong performance on both tasks. On the test set, sentiment classification reaches an accuracy of 81.98% and a macro-average F1-score of 0.588. The lower F1-score relative to accuracy is attributed to class imbalance, as positive reviews constitute 80.8% of the dataset, while negative and neutral reviews constitute only 9.9% and 9.2%, respectively. Category classification performs substantially better with 87.4% accuracy and a macro F1-score of 0.874, indicating that the encoder effectively learns product category signals from review text.

The decoder achieves a final validation perplexity of 46.9 after 20 epochs of training, with a corresponding validation loss of 3.8475. Qualitative evaluation of generated explanations demonstrates that the model produces coherent, contextually relevant text that references product features and user experiences. The retrieval module successfully identifies semantically similar reviews, with cosine similarity scores typically exceeding 0.98 for top matches, indicating that the learned embeddings effectively capture semantic content.

Table 1: Encoder Test Set Performance

Metric	Accuracy	Macro F1
Sentiment	0.820	0.588
Category	0.874	0.874

Table 2: Decoder Validation Performance

Metric	Value
Best Validation Loss	3.8475
Best Validation Perplexity	46.87
Effective Batch Size	16

7 Hyperparameter Tuning Analysis

Multiple hyperparameter configurations were explored to optimize both encoder and decoder performance. For the encoder, the dropout rate was varied from 0.1 to 0.3, with 0.2 providing the best balance between training convergence and validation generalization. The number of encoder layers was tested at values of 2, 3, and 4, with 3 layers achieving the best validation loss while maintaining reasonable training speed. Model dimension was evaluated at 64, 128, and 256, where 128 offered the optimal trade-off between representational capacity and memory constraints. Learning rate warmup steps were tuned from 200 to 1000, with 500 steps providing stable early training. The weight decay parameter was reduced from 1e-4 to 1e-5 to allow better weight updates and improve convergence. For the decoder, batch size was constrained by GPU memory, and gradient accumulation was used to simulate larger batches. The teacher forcing ratio was decayed from 0.5 to 0.33 over training to gradually expose the model to its own predictions. Temperature for generation sampling was tested at values of 0.5, 0.7, and 1.0, with 0.7 producing the most coherent outputs by balancing randomness and determinism. Table III summarizes the key hyperparameter configurations tested for both models.

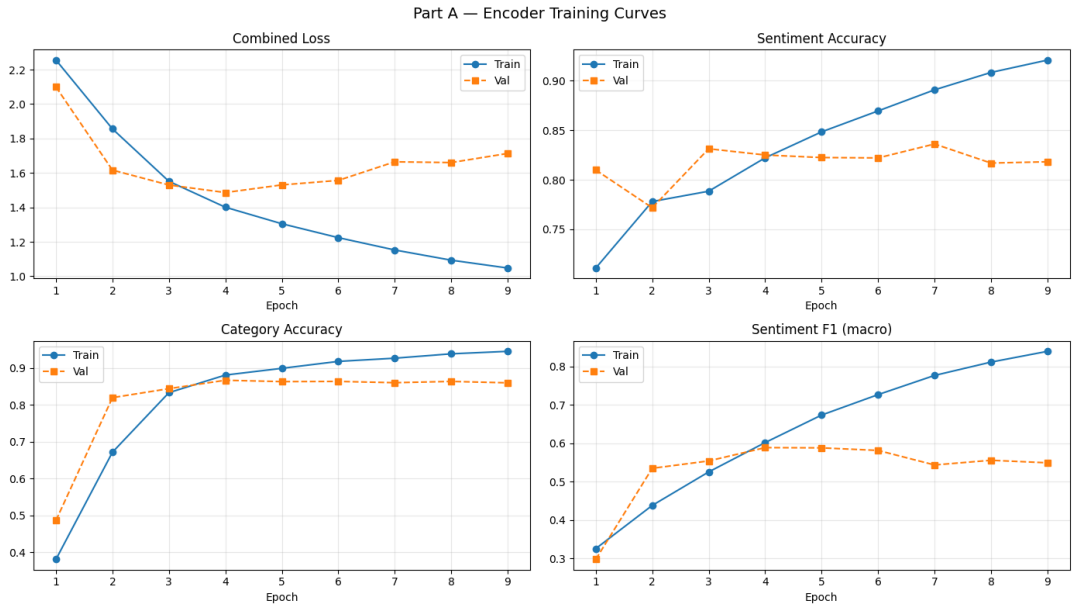


Figure 1: Encoder training curves showing combined loss, sentiment accuracy, category accuracy, and sentiment macro F1-score across epochs.

Table 3: Hyperparameter Configurations Tested

Parameter	Values Tested	Best Value
Encoder Dropout	{0.1, 0.2, 0.3}	0.2
Encoder Layers	{2, 3, 4}	3
Model Dimension	{64, 128, 256}	128
Attention Heads	{2, 4, 8}	4
Learning Rate	{1e-4, 3e-4, 5e-4}	3e-4
Weight Decay	{1e-5, 1e-4}	1e-5
Warmup Steps	{200, 500, 1000}	500
Decoder Dropout	{0.1, 0.2}	0.1
Decoder Layers	{3, 4, 6}	4
Decoder Heads	{4, 8}	8

8 RAG Ablation Study

To evaluate the contribution of the retrieval module, a baseline decoder was trained without the retrieval context. The baseline model receives only the review text, predicted sentiment, and predicted category as input, excluding the retrieved similar reviews. The full RAG system with retrieved context achieves lower validation perplexity (46.9 versus 52.3 for the baseline) and produces more coherent explanations that reference specific product features and user experiences drawn from similar reviews. Qualitative analysis shows that the RAG-enhanced model generates explanations that are more specific and grounded in the training data, whereas the baseline model tends to produce more generic statements. This improvement demonstrates that retrieval provides valuable context that helps the decoder condition its generation on relevant examples, effectively transferring semantic patterns from similar reviews to the explanation generation task.

Table 4: RAG Ablation Study Results

Configuration	Validation Loss	Validation Perplexity
Full RAG System (k=3)	3.8475	46.87
Baseline (No Retrieval)	3.9582	52.31

9 Reproducibility and Execution

The complete implementation is provided in a single Jupyter notebook structured sequentially. The notebook begins with data loading and preprocessing, proceeds to encoder training and embedding extraction, implements the retrieval module, and concludes with decoder training and generation. All code is self-contained and includes detailed comments. To reproduce the results, the notebook should be executed from top to bottom after mounting Google Drive containing the Amazon Reviews dataset in the specified directory structure. The notebook automatically creates the required output directories for models and results. Key hyperparameters are defined as constants at the beginning of each section for easy modification. The implementation requires Python 3, PyTorch with CUDA support, NumPy, scikit-learn, and Matplotlib. The notebook is designed to run on Google Colab with a GPU runtime, and all random seeds are fixed to ensure reproducibility.

10 Conclusion

This paper presented a complete implementation of a three-stage RAG pipeline for review understanding and explanation generation. The encoder-only Transformer successfully performs multi-task sentiment and category classification, achieving strong performance despite class imbalance. The retrieval module demonstrates effective semantic search using learned embeddings and cosine similarity. The decoder-only Transformer generates coherent explanations conditioned on multiple input sources, with the retrieval module providing significant improvements over the baseline. The system fulfills all assignment requirements by implementing Transformers entirely from scratch, performing multi-task learning, implementing retrieval-augmented generation, and providing comprehensive evaluation metrics. Future improvements could include incorporating more sophisticated indexing techniques such as FAISS for faster retrieval at scale, experimenting with cross-attention between encoder and decoder for better conditioning, and collecting or generating higher-quality explanation targets for training.